Problem 16.3.15

Evaluate counterclockwise.

$$\oint_C \tan 2\pi z \; dz, \quad C : |z - 0.2| = 0.2$$

Solution

The function $f(z) = \tan 2\pi z = \frac{\sin 2\pi z}{\cos 2\pi z}$ has simple poles at

$$\cos 2\pi z = 0 \implies 2\pi z = \frac{\pi}{2} + n\pi \implies z = \frac{1}{4} + \frac{n}{2}, \; n \in \mathbb{Z}$$

Only $z_0 = \frac{1}{4}$ lies in C,

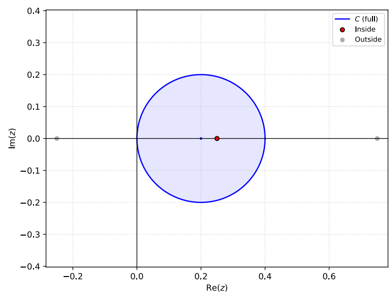

Figure 1: Poles

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{p(z_0)}{q'(z_0)} = \frac{\sin 2\pi z}{-2\pi \sin 2\pi z} = -\frac{1}{2\pi}$$

The integral is then in the form,

$$\oint_C f(z) \; dz = 2\pi i \operatorname*{Res}_{z=z_0} f(z)$$

$$\oint_C \tan 2\pi z \; dz = 2\pi i \operatorname*{Res}_{z=\frac{1}{4}} f(z) = \boxed{-i}$$

Problem 16.3.17

Evaluate counterclockwise.

$$\oint_C \frac{e^z}{\cos z}\, dz, \quad C: \left| z - \frac{\pi i}{2} \right| = 4.5$$

Solution

The function $f(z) = \frac{e^z}{\cos z}$ has simple poles at

$$\cos z = 0 \implies z = \frac{\pi}{2} + n\pi,\ n \in \mathbb{Z}$$

Poles at $z_{1,2} = \pm\frac{\pi}{2}$ lie within the region,

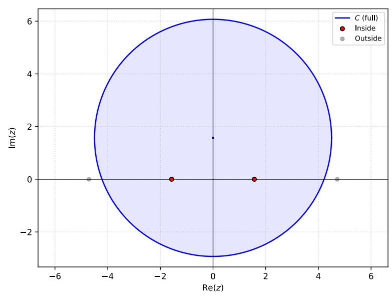

Figure 2: Poles

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{p(z_0)}{q'(z_0)} = -\frac{e^{z_0}}{\sin z_0}$$

$$\operatorname*{Res}_{z=\frac{\pi}{2}} f(z) = -\frac{e^{\frac{\pi}{2}}}{\sin\frac{\pi}{2}} = -e^{\frac{\pi}{2}}$$

$$\operatorname*{Res}_{z=-\frac{\pi}{2}} f(z) = -\frac{e^{-\frac{\pi}{2}}}{\sin -\frac{\pi}{2}} = e^{-\frac{\pi}{2}}$$

The integral is then in the form,

$$\oint_C f(z)\ dz = 2\pi i \sum_{j=1}^{k} \operatorname*{Res}_{z=z_j} f(z)$$

$$\oint_C \frac{e^z}{\cos z}\ dz = 2\pi i \left(-e^{\frac{\pi}{2}} + e^{-\frac{\pi}{2}}\right) = \boxed{-4\pi i \sinh\left(\frac{\pi}{2}\right)}$$

Problem 16.3.19

Evaluate counterclockwise.

$$\oint_C \frac{\sinh z}{2z - i}\, dz, \quad C : |z - 2i| = 2$$

Solution

The function $f(z) = \frac{\sinh z}{2z-i}$ has simple poles at

$$2z - i = 0 \implies z = \frac{i}{2}$$

The pole at $z_0 = \frac{i}{2}$ lies within the region,

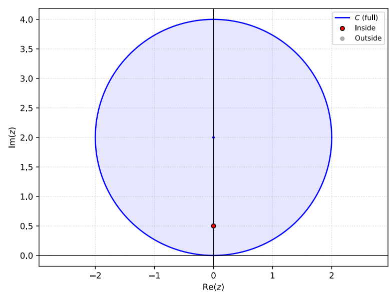

Figure 3: Poles

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{p(z_0)}{q'(z_0)} = \frac{1}{2}\sinh z_0$$

$$\operatorname*{Res}_{z=\frac{i}{2}} f(z) = \frac{1}{2}\sinh\frac{i}{2} = \frac{i}{2}\sin\frac{1}{2}$$

The integral is then in the form,

$$\oint_C f(z)\, dz = 2\pi i \operatorname*{Res}_{z=z_0} f(z)$$

$$\oint_C \frac{\sinh z}{2z - i}\, dz = 2\pi i\left(\frac{i}{2}\sin\frac{1}{2}\right) = \boxed{-\pi \sin\frac{i}{2}}$$

Problem 16.3.21

Evaluate counterclockwise.

$$\oint_C \frac{\cos \pi z}{z^5}\, dz, \quad C : |z| = \frac{1}{2}$$

Solution

The function $f(z) = \frac{\cos \pi z}{z^5}$ has a 5th-order pole at

$$z^5 = 0 \implies z = 0$$

The pole at $z_0 = 0$ lies within the region,

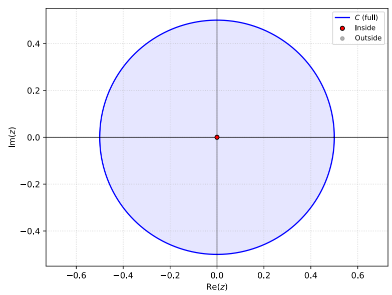

Figure 4: Poles

$$\operatorname*{Res}_{z=0} f(z) = \frac{1}{4!} \lim_{z\to 0} \left\{ \frac{d^4}{dz^4} \left[z^4 f(z) \right] \right\} = \frac{1}{24} \frac{d^4}{dz^4} \left[\cos \pi z \right] \Big|_{z=0} = \frac{\pi^4}{24} \cos \pi z \Big|_{z=0} = \frac{\pi^4}{24}$$

The integral is then in the form,

$$\oint_C f(z)\, dz = 2\pi i \operatorname*{Res}_{z=z_0} f(z)$$

$$\oint_C \frac{\cos \pi z}{z^5}\, dz = 2\pi i \left(\frac{\pi^4}{24} \right) = \boxed{\frac{i\pi^5}{12}}$$

Problem 16.4.1

Evaluate.

$$\int_0^{\pi} \frac{2}{k - \cos\theta}\, d\theta$$

Solution

The integral can be converted to a complex integral by substituting $z = e^{i\theta}$

$$\int_0^{\pi} \frac{2}{k - \cos\theta}\, d\theta = \int_0^{2\pi} \frac{1}{k - \cos\theta}\, d\theta = \oint_C \frac{2}{k - \frac{1}{2}\left(z + \frac{1}{z}\right)} \frac{dz}{iz} = 2i \oint_C \frac{1}{z^2 - 2kz + 1}\, dz$$

Where C is the the unit circle in the complex plane.

$f(z) = \frac{1}{z^2 - 2kz + 1}$ has simple poles where

$$z^2 - 2kz + 1 = 0 \implies z = \frac{2k \pm \sqrt{4k^2 - 4}}{2} = k \pm \sqrt{k^2 - 1}$$

$$z_1 = k - \sqrt{k^2 - 1} \quad z_2 = k + \sqrt{k^2 - 1}$$

$$\operatorname*{Res}_{z=z_0} f(z) = \frac{1}{2z_0 - 2k} = \frac{1}{2(z_0 - k)}$$

$$\operatorname*{Res}_{z=z_{1,2}} f(z) = \frac{1}{2(k \pm \sqrt{k^2 - 1} - k)} = \pm \frac{1}{2\sqrt{k^2 - 1}}$$

Assuming $k > 1$, only z_1 lies within C,

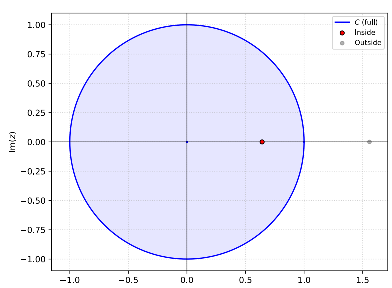

Figure 5: Poles (e.g. $k = 1.1$)

$$\oint_C f(z)\, dz = 2\pi i \operatorname*{Res}_{z=z_0} f(z)$$

$$\oint_C \frac{1}{z^2 - 2kz + 1} = 2\pi i \left(-\frac{1}{2\sqrt{k^2 - 1}} \right) \quad = -\frac{\pi i}{\sqrt{k^2 - 1}}$$

$$\int_0^{\pi} \frac{2}{k - \cos\theta}\, d\theta = 2i \left(-\frac{\pi i}{\sqrt{k^2 - 1}} \right)$$

$$= \boxed{\frac{2\pi}{\sqrt{k^2 - 1}}}$$

Problem 16.4.7

Evaluate.

$$\int_0^{2\pi} \frac{a}{a - \sin\theta}\, d\theta$$

Solution

Problem 16.4.9

Evaluate.

$$\int_0^{2\pi} \frac{\cos\theta}{13 - 12\cos 2\theta}\ d\theta$$

Solution

Problem 16.4.11

Evaluate.

$$\int_{-\infty}^{\infty} \frac{1}{\left(1+x^2\right)^2}\ dx$$

Solution

Problem 16.4.15

Evaluate.

$$\int_{-\infty}^{\infty} \frac{x^2}{x^6+1}\ dx$$

Solution

Problem 16.4.17

Evaluate.

$$\int_{-\infty}^{\infty} \frac{\sin 3x}{x^4+1}\ dx$$

Solution

Problem 16.4.19

Evaluate.

$$\int_{-\infty}^{\infty} \frac{1}{x^4 - 1} \, dx$$

Solution

Problem 16.4.21

Evaluate.

$$\int_{-\infty}^{\infty} \frac{\sin x}{(x-1)\left(x^2+4\right)}\ dx$$

Solution

Problem 16.4.25

Find the Cauchy principal value.

$$\int_{-\infty}^{\infty} \frac{x+5}{x^3 - x}\, dx$$